# ALEXANDRA BREMERS

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I study how machines can collaborate with people in the physical world. My approach combines design activities with systems prototyping and user studies. Prior to joining Cornell, I studied and developed future AI and AR human-machine interfaces at Jaguar Land Rover.

## EDUCATION

2020- **Cornell University**, Cornell Tech, New York, NY, USA
Ph.D. in Information Science
Specialization: Human-Computer Interaction
Minor: Computer Science
Committee: Wendy Ju (chair), François Guimbretière, Stephen Marschner
Expected graduation date: August 2025

2016-2018 **Utrecht University**, Utrecht, The Netherlands
M.Sc. in Artificial Intelligence
Track: Cognitive Processing
Thesis: *Perception of Perspective in Augmented Reality Head-Up Displays*
Advisors: Chris Janssen, Ignace Hooge, Valerian Meijering

2013-2016 **Eindhoven University of Technology**, Eindhoven, The Netherlands
B.Sc. in Industrial Design
Graduation Project: *How to embroider a radio*
Advisors: Oscar Tomico, Saskia Bakker

2007-2013 **Stedelijk Gymnasium Arnhem**, Arnhem, The Netherlands
Gymnasium (Grammar School)

## INDUSTRY RESEARCH EXPERIENCE

2023 (summer) **Associate Principal**, Accenture Labs (R&D)
San Francisco, CA, USA

2021 (summer) **Research Intern**, Toyota Research Institute
Los Altos, CA, USA (remote)

2017–2020 **Human-Machine Interface Researcher**, Jaguar Land Rover
National Automotive Innovation Center, Coventry, United Kingdom

## ACADEMIC RESEARCH EXPERIENCE

2020– **Graduate Research Assistant**, Future Automation Research Lab
Cornell Tech, Cornell University, New York, NY, USA
Collaborations with Amada AI Innovation Laboratory Ltd., Accenture Labs, Toyota Research Institute and Nissan Motors.

2022 (summer) **Research Visitor**, User Interfaces Group, Department of Engineering
Aalto University, Espoo, Finland

2021 (fall) **Research Visitor**, Robots in Groups Lab
Cornell University, Ithaca, NY, USA

2017 (summer) **Research Intern**, NTU IoX Center
National Taiwan University, Taipei, Taiwan

## PUBLICATIONS

Journal Articles

Nermin Caber, Jiaming Liang, Bashar I. Ahmad, Simon Godsill, **Alexandra Bremers**, Philip Thomas, David Oxtoby, Lee Skrypchuk. "Driver Profiling and Bayesian Workload Estimation for Adaptive In-Vehicle HMI". In: *IEEE Transactions on Intelligent Vehicles*. 2023. DOI: https://doi.org/10.48550/arXiv.2303.14720.

**Alexandra W.D. Bremers**, Ali Özgür Yöntem, Kun Li, Daping Chu, Valerian Meijering, Christian P. Janssen. "Perception of Perspective in Augmented Reality Head-Up Displays". In: *International Journal of Human-Computer Studies)* (2021), p. 102693. ISSN: 1071-5819. DOI: https://doi.org/10.1016/j.ijhcs.2021.102693. URL: https://www.sciencedirect.com/science/article/pii/S1071581921001117.

Full Conference Papers

**Alexandra Bremers**, Maria Teresa Parreira, Xy Fang, Natalie Friedman, Adolfo Ramirez-Aristizabal, Alexandria Pabst, Mirjana Spasojevic, Mike Kuniavsky, Wendy Ju. "The Bystander Affect Detection (BAD) dataset for failure detection in HRI". In: *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2023. DOI: https://doi.org/10.48550/arXiv.2303.04835.

Avital Dell'Ariccia, **Alexandra Bremers**, Johan Michalove, Wendy Ju. "How to make people think you're thinking if you're a drawing robot". In: *Proceedings of the 17th ACM Conference on Human-Robot Interaction*. 2022. DOI: https://doi.org/10.1109/HRI53351.2022.9889638. URL: https://ieeexplore.ieee.org/abstract/document/9889638.

David Goedicke, **Alexandra Bremers**, Sam Lee, Fanjun Bu, Hiroshi Yasuda, Wendy Ju. "XR-OOM: MiXed Reality Driving Simulation With Real Cars". In: *Proceedings of the ACM International Conference on Human Computer Interaction (CHI)*. 2022. DOI: https://doi.org/10.1145/3491102.3517704. URL: https://dl.acm.org/doi/abs/10.1145/3491102.3517704.

Under Review

**Alexandra Bremers**, Alexandria Pabst, Maria Teresa Parreira, Wendy Ju. "Using Social Cues to Recognize Task Failures for HRI: A Review of Current Research and Future Directions". In: *arXiv:2301.11972*. 2023. DOI: https://doi.org/10.48550/arXiv.2301.11972.

Natalie Friedman, Asmita Mehta, Kari Love, **Alexandra W.D. Bremers**, Awsaf Ahmed, Wendy Ju. "A utility belt for an agricultural robot: reflection-in-action for applied design research". In: *(under review)*. 2023.

Peer-Reviewed Abstracts

**Alexandra Bremers**, Maria Teresa Parreira, Wendy Ju. "Understanding Bystander Facial Responses to Robot Task Failures with the BAD Dataset". In: *HRI Workshop Position: The Imperfectly Relatable Robot*. 2023.

**Alexandra Bremers**. "A computer that sketches along with you". In: *Creativity and Cognition*. C&C '22. Venice, Italy, 2022. DOI: https://doi.org/10.1145/3527927.3533732. URL: https://dl.acm.org/doi/abs/10.1145/3527927.3533732.

**Alexandra Bremers**. "How can a robot help people draw?" In: *Companion Publication of the 2021 ACM Designing Interactive Systems Conference*. DIS '22 Companion. Virtual Event: Association for Computing Machinery, 2022. DOI: https://doi.org/10.1145/3532107.3532876. URL: https://dl.acm.org/doi/abs/10.1145/3532107.3532876.

Avital Dell'Ariccia, **Alexandra Bremers**, Wen-Ying Lee, Wendy Ju. "Work in Progress: "Ah! He wants to win!": Social responses to playing Tic-Tac-Toe against a physical drawing robot". In: *Adjunct Proceedings of the 2022 ACM International Conference on Tangible and Embodied Interaction*. 2022. DOI: https://doi.org/10.1145/3490149.3505571. URL: https://dl.acm.org/doi/abs/10.1145/3490149.3505571.

Natalie Friedman, Kari Love, **Alexandra Bremers**, AJ Parry, Ray LC, Bolor Amgalan, Jen Liu, Wendy Ju. "Workshop: Designing Functional Clothing for Human-robot Interaction". In: *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*. 2021, pp. 703–705. DOI: https://doi.org/10.1145/3434074.3444870. URL: https://dl.acm.org/doi/abs/10.1145/3434074.3444870.

David Goedicke, **Alexandra Bremers**, Hiroshi Yasuda, Wendy Ju. "Work in Progress: XR-OOM: Mixing Driving Environments with Virtual Objects Safely". In: *Adjunct Proceedings of the 2021 ACM International Conference on Automotive Interfaces and Vehicular Applications*. 2021. DOI: https://doi.org/10.1145/3473682.3480266. URL: https://dl.acm.org/doi/abs/10.1145/3473682.3480266.

## GRANTS, HONOURS & AWARDS

**Fellowships**

Google Fellowship Departmental Nominee
*Date:* 2022
Selected candidate by Cornell University Bowers CIS - Information Science.

Royal Commission of 1851 Industrial Fellowship
*Amount:* £90,000
*Date:* 2020
Fellowship to pursue a PhD at the University of Cambridge Department of Engineering. Declined in order to accept Cornell offer.

**Grants**

Cornell Graduate School Conference Grant
*Amount:* $700
*Date:* 2023

Finnish Center for AI (HIIT/FCAI) Travel Support Grant
*Amount:* €3,500
*Date:* 2022

ACM Creativity & Cognition Graduate Symposium NSF Student Travel Grant
*Amount:* $1,900
*Date:* 2022

Cornell Graduate School Conference Grant
*Amount:* $700
*Date:* 2022

Cornell Graduate School Conference Grant
*Amount:* $675
*Date:* 2021

Cornell Tech Backslash Art Microgrant: **Plotting with Conductive Inks.**
*Amount:* $500
*Date:* 2021

**Honours & Awards**

CRA-WP Grad Cohort for Women
Invited to conference in San Francisco
*Date:* 2023

TATA Innovista Global Finalist & JLR Innovista Global Finalist
*Date:* 2019

Oxford-Cambridge Rising Women in Science and Engineering
Invited to course at Oxford Saïd Business School and Cambridge Madingley Hall
*Date:* 2018

Bruynzeel Storage Systems, Designing the Office Competition, 2nd Student Prize
*Value:* €500
*Date:* 2015

Schoolkrantawards, Best School Paper of The Netherlands (Best Content)
*Value:* €250
*Date:* 2012

Schoolkrantawards, Best School Paper of The Netherlands (Best Layout)
*Value:* €250
*Date:* 2012

Lions Club Arnhem Arenacum & Stedelijk Gymnasium Arnhem, Best Report
*Value:* €200
*Date:* 2012

## TALKS

Invited talks

Invited Speaker, "Interaction Intelligence for HRI". Accenture Labs California (online). With Wendy Ju. October 21st, 2022.

Invited Speaker, Cornell Information Science Seminar. April 2022.

Invited participant and panelist, "How can models inform design?". Computational Models of Human-Automated Vehicle Interaction Seminar. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany. March 6-11, 2022.

Invited Speaker, "After RisingWISE". RisingWISE reunion event by University of Oxford and University of Cambridge. Woburn Safari Park Lodge, United Kingdom. September 25th, 2021.

Guest Speaker, "XR-OOM: miXed Reality On-rOad siMulator". XR-Monthly by the XR-Collective at Cornell Tech (online). With David Goedicke. February 19th, 2021.

Invited Participant, "RisingWISE". Event series by University of Oxford and University of Cambridge to connect women in STEM research. Held between Saïd Business School Oxford and Madingley Hall Cambridge, United Kingdom. November 2018 - January 2019.

Other

Presenter. Cornell Tech outreach (online). July 2022.

Presenter. Computing and Information Science Research Night. Gates Hall, Cornell University, Ithaca, NY. November 2, 2021.

## ADVISING

2021– **Internship Advisor**, Cornell Tech, Cornell University, New York, NY

2023 Itay Grinberg: (M, Technion)

2022 Cooper Murr: *Translating digital algorithms to paper*. (UG, Cornell)

2022 Daniel Zhou: *Exploring HAI interaction: calculator*. (UG, Cornell)

2022 Tobias Weinberg: *Drawing together*. (UG, Technion)

2022 Mary Lee: *Bad Robots*. (High School)

2021 Avital Dell'Ariccia: *Robotic Tic-Tac-Toe*. (UG, Technion, Israel)

- Work-in-Progress paper accepted at TEI 2022.
- Video accepted at HRI 2022.
- Awarded Gary Marshden Travel Grant to HRI and TEI.

## TEACHING

2021– **Teaching Assistant**, Cornell Tech, Cornell University, New York, NY, USA

2022 (F) CS5424 Developing and Designing Interactive Devices

2021 (F) CS5424 Developing and Designing Interactive Devices

2021 (S) CS5678 Topics in Mixed Reality

## DESIGN & ART EXPERIENCE

2015 (fall) **Speculative Design Intern**
Next Nature Network, Amsterdam, The Netherlands

2013–2015 **Volunteer at Interactive Art Exhibitions**
Dutch Design Week / Fiber Festival / GLOW Festival, The Netherlands

## EDUCATION (OTHER)

2019 **Warwick University**, Coventry, United Kingdom
M.Eng. Module (5 CATS Credits), Human-Technology Interaction for Automotive

2010 **Radboud University**, Nijmegen, The Netherlands
Non-degree module, Designing and Making Web-Based Software

## SERVICE

Departmental Service

2022-23

- Task Force Design+Technology Center and Initiative, Cornell

2021-22

- Task Force Design+Technology Center and Initiative, Cornell
- PhD admissions reviewer and interviewer

2020-21

- Task Force Design+Technology Center and Initiative, Cornell
- PhD admissions reviewer and interviewer

Academic Service

Active service in:

ACM SIGGRAPH (Special Interest Group on Computer Graphics and Interactive Techniques)

2023 Demo Volunteer, Blasts from the Past Exhibit

ACM DIS (Designing Interactive Systems)

2021 Social Media Chair

ACM AutoUI (Automotive User Interfaces and Interactive Vehicular Applications)

2020 Moderator

2019 Visual Design Chair

Peer Reviewer for:

2023 ACM THRI

2022 ACM CHI

2021 ACM AutoUI

2021 ACM IMWUT

2021 ACM DIS

2021 IEEE IV

2020 ACM CHI

2019 ACM Auto-UI

## ASSOCIATIONS

Association for Computing Machinery

Optical Society of America

## PROGRAMMING LANGUAGES

Proficient: Python, R
Experienced: C++ (Arduino), C# (Unity), Java (Processing)